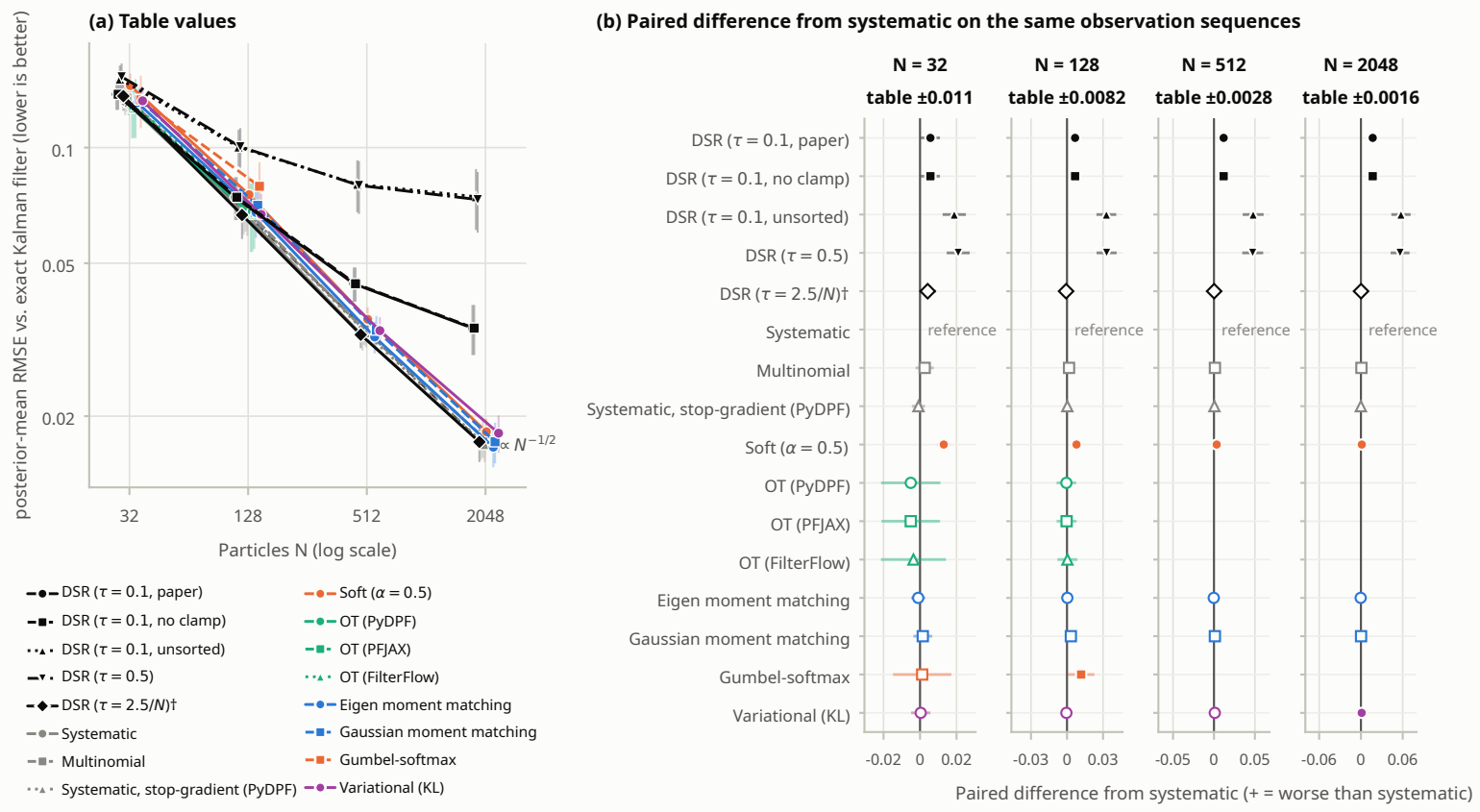

(a) Table values
posterior-mean RMSE vs. exact Kalman filter (lower is better)
0.1
0.05
0.02
∝ $N^{-1/2}$
32
128
512
2048
Particles N (log scale)
DSR (τ = 0.1, paper)
DSR (τ = 0.1, no clamp)
DSR (τ = 0.1, unsorted)
DSR (τ = 0.5)
DSR (τ = 2.5/N)†
Systematic
Multinomial
Systematic, stop-gradient (PyDPF)
Soft (α = 0.5)
OT (PyDPF)
OT (PFJAX)
OT (FilterFlow)
Eigen moment matching
Gaussian moment matching
Gumbel-softmax
Variational (KL)
(b) Paired difference from systematic on the same observation sequences
N = 32
table ±0.011
N = 128
table ±0.0082
N = 512
table ±0.0028
N = 2048
table ±0.0016
DSR (τ = 0.1, paper)
DSR (τ = 0.1, no clamp)
DSR (τ = 0.1, unsorted)
DSR (τ = 0.5)
DSR (τ = 2.5/N)†
Systematic
Multinomial
Systematic, stop-gradient (PyDPF)
Soft (α = 0.5)
OT (PyDPF)
OT (PFJAX)
OT (FilterFlow)
Eigen moment matching
Gaussian moment matching
Gumbel-softmax
Variational (KL)
reference
-0.02 0 0.02
-0.03 0 0.03
-0.05 0 0.05
-0.06 0 0.06
Paired difference from systematic (+ = worse than systematic)